

Problem 29.2: Electron-Positron Annihilation into Hadrons

Source: M. D. Schwartz, *Quantum Field Theory and the Standard Model*, Problem 29.2.

$e^+e^- \rightarrow \text{hadrons}$ in the Standard Model.

1. Calculate the rate for the total cross section $\sigma_{\text{tot}}(e^+e^- \rightarrow \text{hadrons})$ in the Standard Model at tree-level including both Z -boson and photon contributions and their interference. The contribution using photons alone was calculated in Section 26.3.
2. Calculate σ_{tot} at 1-loop.
3. Plot the total cross section as a function of center-of-mass energy showing separately the photon contribution, the Z -boson contribution, and their sum. Plot also the sum ignoring interference between the Z -boson and photon contributions. When can interference be ignored?

Solution. For the $e^+e^- \rightarrow \gamma^* \rightarrow \text{hadrons}$, the relevant sector of the electroweak Lagrangian is

$$\mathcal{L} = \bar{\psi}_e(i\not{\partial} - m_e)\psi_e + \sum_i \bar{\psi}_i(i\not{\partial} - m_i)\psi_i - e\bar{\psi}_e\not{A}\psi_e + e \sum_i Q_i \bar{\psi}_i\not{A}\psi_i, \quad (1.1)$$

where i denotes the up, down, strange, etc. We can read off the scattering amplitude directly for a particular pair of quark and anti-quark, which is

$$i\mathcal{M}_\gamma^i = (ie)^2 Q_i [\bar{v}(p_2)\gamma^\mu u(p_1)] \frac{-i}{s} [\bar{u}_i(p_3)\gamma_\mu v_i(p_4)] = iQ_i \mathcal{M}_0, \quad (1.2)$$

where $\mathcal{M}_0$ is the scattering amplitude of $e^-e^+ \rightarrow \mu^-\mu^+$ at tree-level. Note that $Q_{u,c,t} = \frac{2}{3}$ and $Q_{d,s,b} = -\frac{1}{3}$.

The Z boson coupling to a general quark field is given by

$$\frac{ie}{\sin\theta_w \cos\theta_w} \gamma^\mu \left(T^3 \frac{1-\gamma^5}{2} - Q_i \sin^2\theta_w \right) = \frac{ie}{4 \sin\theta_w \cos\theta_w} \gamma^\mu (\xi_i \gamma^5 - 4Q_i \sin^2\theta_w - \xi_i), \quad (1.3)$$

where

$$\xi_i = \begin{cases} -1, & i = u, c, t, \\ +1, & i = d, s, b. \end{cases} \quad (1.4)$$

Defining

$$a_i = 4|Q_i| \sin^2\theta_w - 1, \quad (1.5)$$

the quark vertex can equivalently be written as

$$\frac{ie}{4 \sin\theta_w \cos\theta_w} \xi_i \gamma^\mu (\gamma^5 + a_i). \quad (1.6)$$

Therefore, the Z -exchange amplitude is

$$i\mathcal{M}_Z^i = \left(\frac{ie}{4 \sin \theta_w \cos \theta_w} \right)^2 [\bar{v}(p_2) \gamma^\mu (\gamma^5 + a) u(p_1)] \frac{-i}{s - m_Z^2} [\xi_i \bar{u}(p_3) \gamma_\mu (\gamma^5 + a_i) v(p_4)], \quad (1.7)$$

where

$$a = 4 \sin^2 \theta_w - 1. \quad (1.8)$$

The initial-state spin sum is

$$\begin{aligned} & \sum_{\text{spins}} [\bar{v}(p_2) \gamma^\mu (\gamma^5 + a) u(p_1)] [\bar{u}(p_1) \gamma^\nu (\gamma^5 + a) v(p_2)] \\ &= \text{Tr} \left[\not{p}_2 \gamma^\mu (\gamma^5 + a) \not{p}_1 \gamma^\nu (\gamma^5 + a) \right] \\ &= 4(a^2 + 1) (p_1^\mu p_2^\nu + p_1^\nu p_2^\mu - p_1 \cdot p_2 \eta^{\mu\nu}) - 8ia p_{1\alpha} p_{2\beta} \epsilon^{\alpha\mu\beta\nu}. \end{aligned} \quad (1.9)$$

Similarly, since $\xi_i^2 = 1$, the final-state spin sum is

$$4(a_i^2 + 1) (p_{3\mu} p_{4\nu} + p_{3\nu} p_{4\mu} - p_3 \cdot p_4 \eta_{\mu\nu}) + 8ia_i p_{3\alpha} p_{4\beta} \epsilon_\mu^{\alpha\beta\nu}. \quad (1.10)$$

Thus,

$$\begin{aligned} \sum_{\text{spins}} |\mathcal{M}_Z^i|^2 &= \left(\frac{e}{4 \sin \theta_w \cos \theta_w} \right)^4 \frac{1}{(s - m_Z^2)^2} \\ &\times \left[4(a^2 + 1) (p_1^\mu p_2^\nu + p_1^\nu p_2^\mu - p_1 \cdot p_2 \eta^{\mu\nu}) - 8ia p_{1\alpha} p_{2\beta} \epsilon^{\alpha\mu\beta\nu} \right] \\ &\times \left[4(a_i^2 + 1) (p_{3\mu} p_{4\nu} + p_{3\nu} p_{4\mu} - p_3 \cdot p_4 \eta_{\mu\nu}) + 8ia_i p_{3\alpha} p_{4\beta} \epsilon_\mu^{\alpha\beta\nu} \right]. \end{aligned} \quad (1.11)$$

The contraction between a symmetric tensor and an antisymmetric tensor vanishes. Contracting the two symmetric terms and the two epsilon tensors therefore gives

$$\begin{aligned} \sum_{\text{spins}} |\mathcal{M}_Z^i|^2 &= \frac{e^4}{8 \sin^4 \theta_w \cos^4 \theta_w (s - m_Z^2)^2} \\ &\times \left[((a^2 + 1)(a_i^2 + 1) - 4aa_i)(p_1 \cdot p_3)(p_2 \cdot p_4) \right. \\ &\quad \left. + ((a^2 + 1)(a_i^2 + 1) + 4aa_i)(p_1 \cdot p_4)(p_2 \cdot p_3) \right]. \end{aligned} \quad (1.12)$$

For massless external particles,

$$p_1 \cdot p_3 = p_2 \cdot p_4 = -\frac{t}{2}, \quad p_1 \cdot p_4 = p_2 \cdot p_3 = -\frac{u}{2}. \quad (1.13)$$

Hence,

$$\begin{aligned} \sum_{\text{spins}} |\mathcal{M}_Z^i|^2 &= \frac{e^4}{32 \sin^4 \theta_w \cos^4 \theta_w (s - m_Z^2)^2} \\ &\times \left[((a^2 + 1)(a_i^2 + 1) - 4aa_i)t^2 + ((a^2 + 1)(a_i^2 + 1) + 4aa_i)u^2 \right]. \end{aligned} \quad (1.14)$$

For the photon-- Z interference term, the required mixed traces are

$$\begin{aligned}
\text{Tr} \left[\not{\epsilon}_2 \gamma^\mu \not{\epsilon}_1 \gamma^\nu (\gamma^5 + a) \right] &= 4a (p_1^\mu p_2^\nu + p_1^\nu p_2^\mu - p_1 \cdot p_2 \eta^{\mu\nu}) - 4i p_{1\alpha} p_{2\beta} \epsilon^{\alpha\mu\beta\nu}, \\
\text{Tr} \left[\not{\epsilon}_3 \gamma_\mu \not{\epsilon}_4 \gamma_\nu (\xi_i \gamma^5 + \xi_i a_i) \right] &= 4\xi_i a_i (p_{3\mu} p_{4\nu} + p_{3\nu} p_{4\mu} - p_3 \cdot p_4 \eta_{\mu\nu}) \\
&\quad + 4i \xi_i p_{3\alpha} p_{4\beta} \epsilon^\alpha{}_\mu{}^\beta{}_\nu.
\end{aligned} \tag{1.15}$$

Therefore,

$$\begin{aligned}
2 \text{Re} \sum_{\text{spins}} \mathcal{M}_\gamma^i \mathcal{M}_Z^{i\dagger} &= \frac{e^4 Q_i \xi_i}{\sin^2 \theta_w \cos^2 \theta_w s(s - m_Z^2)} \\
&\quad \times [(aa_i - 1)t^2 + (aa_i + 1)u^2].
\end{aligned} \tag{1.16}$$

Putting the photon, Z , and interference contributions together and averaging over the initial spins gives

$$\begin{aligned}
\frac{1}{4} \sum_{\text{spin}} |\mathcal{M}^i|^2 &= \frac{e^4 Q_i^2}{s^2} (t^2 + u^2) \\
&\quad + \frac{e^4}{128 \sin^4 \theta_w \cos^4 \theta_w (s - m_Z^2)^2} [((a^2 + 1)(a_i^2 + 1) - 4aa_i)t^2 \\
&\quad + ((a^2 + 1)(a_i^2 + 1) + 4aa_i)u^2] \\
&\quad + \frac{e^4 Q_i \xi_i}{4 \sin^2 \theta_w \cos^2 \theta_w s(s - m_Z^2)} [(aa_i - 1)t^2 + (aa_i + 1)u^2].
\end{aligned} \tag{1.17}$$

Then, we sum over all flavours, which can be simplified as summing over the up and down flavour times three in our massless approximation. That is

$$\frac{1}{4} \sum_i \sum_{\text{spin}} |\mathcal{M}^i|^2 = \frac{3}{4} \sum_{\text{spin}} |\mathcal{M}^u|^2 + |\mathcal{M}^d|^2. \tag{1.18}$$

Using

$$Q_u = \frac{2}{3}, \quad Q_d = -\frac{1}{3}, \quad \xi_u = -1, \quad \xi_d = 1, \tag{1.19}$$

and

$$a_u = \frac{8}{3} \sin^2 \theta_w - 1, \quad a_d = \frac{4}{3} \sin^2 \theta_w - 1, \tag{1.20}$$

the required sums are

$$\begin{aligned}
3 \sum_{i=u,d} Q_i^2 &= \frac{5}{3}, \\
3 \sum_{i=u,d} [(a^2 + 1)(a_i^2 + 1) - 4aa_i] &= \frac{32}{3} \sin^4 \theta_w (40 \sin^4 \theta_w - 56 \sin^2 \theta_w + 23), \\
3 \sum_{i=u,d} [(a^2 + 1)(a_i^2 + 1) + 4aa_i] &= \frac{16}{3} (80 \sin^8 \theta_w - 112 \sin^6 \theta_w + 118 \sin^4 \theta_w - 54 \sin^2 \theta_w + 9), \\
3 \sum_{i=u,d} Q_i \xi_i (aa_i - 1) &= \frac{8}{3} \sin^2 \theta_w (7 - 10 \sin^2 \theta_w), \\
3 \sum_{i=u,d} Q_i \xi_i (aa_i + 1) &= -\frac{2}{3} (40 \sin^4 \theta_w - 28 \sin^2 \theta_w + 9).
\end{aligned} \tag{1.21}$$

Therefore,

$$\begin{aligned}
\frac{1}{4} \sum_i \sum_{\text{spins}} |\mathcal{M}^i|^2 &= \frac{10e^4}{3s^2} (t^2 + u^2) \\
&+ \frac{e^4}{24 \sin^4 \theta_w \cos^4 \theta_w (s - m_Z^2)^2} \left[2 \sin^4 \theta_w (40 \sin^4 \theta_w - 56 \sin^2 \theta_w + 23) t^2 \right. \\
&+ \left. (80 \sin^8 \theta_w - 112 \sin^6 \theta_w + 118 \sin^4 \theta_w - 54 \sin^2 \theta_w + 9) u^2 \right] \\
&+ \frac{e^4}{6 \sin^2 \theta_w \cos^2 \theta_w s (s - m_Z^2)} \left[4 \sin^2 \theta_w (7 - 10 \sin^2 \theta_w) t^2 \right. \\
&- \left. (40 \sin^4 \theta_w - 28 \sin^2 \theta_w + 9) u^2 \right].
\end{aligned} \tag{1.22}$$

One should also not forget to sum over all three outgoing colour states, namely $r\bar{r}$, $g\bar{g}$, and $b\bar{b}$, which gives another factor of 3. Therefore, the spin-averaged squared amplitude, summed over flavours and colours, is

$$\begin{aligned}
\overline{|\mathcal{M}|^2} &= \frac{10e^4}{s^2} (t^2 + u^2) \\
&+ \frac{e^4}{8 \sin^4 \theta_w \cos^4 \theta_w (s - m_Z^2)^2} \left[2 \sin^4 \theta_w (40 \sin^4 \theta_w - 56 \sin^2 \theta_w + 23) t^2 \right. \\
&+ \left. (80 \sin^8 \theta_w - 112 \sin^6 \theta_w + 118 \sin^4 \theta_w - 54 \sin^2 \theta_w + 9) u^2 \right] \\
&+ \frac{e^4}{2 \sin^2 \theta_w \cos^2 \theta_w s (s - m_Z^2)} \left[4 \sin^2 \theta_w (7 - 10 \sin^2 \theta_w) t^2 \right. \\
&- \left. (40 \sin^4 \theta_w - 28 \sin^2 \theta_w + 9) u^2 \right].
\end{aligned} \tag{1.23}$$

In the centre-of-mass frame, we have

$$t = -\frac{s}{2}(1 - \cos \theta), \quad u = -\frac{s}{2}(1 + \cos \theta). \tag{1.24}$$

For massless two-body scattering,

$$\frac{d\sigma}{d \cos \theta} = \frac{1}{32\pi s} \overline{|\mathcal{M}|^2}. \tag{1.25}$$

Substituting t and u , we obtain

$$\begin{aligned}
\frac{d\sigma}{d \cos \theta} &= \frac{5e^4}{32\pi s} (1 + \cos^2 \theta) \\
&+ \frac{e^4 s}{1024\pi \sin^4 \theta_w \cos^4 \theta_w (s - m_Z^2)^2} \left[2 \sin^4 \theta_w (40 \sin^4 \theta_w - 56 \sin^2 \theta_w + 23) (1 - \cos \theta)^2 \right. \\
&+ \left. (80 \sin^8 \theta_w - 112 \sin^6 \theta_w + 118 \sin^4 \theta_w - 54 \sin^2 \theta_w + 9) (1 + \cos \theta)^2 \right] \\
&+ \frac{e^4}{256\pi \sin^2 \theta_w \cos^2 \theta_w (s - m_Z^2)} \left[4 \sin^2 \theta_w (7 - 10 \sin^2 \theta_w) (1 - \cos \theta)^2 \right. \\
&- \left. (40 \sin^4 \theta_w - 28 \sin^2 \theta_w + 9) (1 + \cos \theta)^2 \right].
\end{aligned} \tag{1.26}$$

Using

$$\int_{-1}^1 (1 \pm \cos \theta)^2 d \cos \theta = \frac{8}{3}, \tag{1.27}$$

the total cross section is

$$\begin{aligned}\sigma_{\text{tot}}(s) = & \frac{5e^4}{12\pi s} \\ & + \frac{e^4 s}{384\pi \sin^4 \theta_w \cos^4 \theta_w (s - m_Z^2)^2} \left[2 \sin^4 \theta_w (40 \sin^4 \theta_w - 56 \sin^2 \theta_w + 23) \right. \\ & + 80 \sin^8 \theta_w - 112 \sin^6 \theta_w + 118 \sin^4 \theta_w - 54 \sin^2 \theta_w + 9 \left. \right] \\ & + \frac{e^4}{96\pi \sin^2 \theta_w \cos^2 \theta_w (s - m_Z^2)} \left[4 \sin^2 \theta_w (7 - 10 \sin^2 \theta_w) \right. \\ & - (40 \sin^4 \theta_w - 28 \sin^2 \theta_w + 9) \left. \right].\end{aligned}\tag{1.28}$$

Taking

$$\alpha = \frac{e^2}{4\pi} = \frac{1}{128}, \quad \sin^2 \theta_w = 0.231, \quad m_Z = 91.2 \text{ GeV},\tag{1.29}$$

and using $1 \text{ GeV}^{-2} = 3.894 \times 10^8 \text{ pb}$, this becomes

$$\sigma_{\text{tot}}(s) = \left[\frac{4.98 \times 10^5}{s} + \frac{2.93 \times 10^5 s}{(s - 8317)^2} - \frac{2.33 \times 10^4}{s - 8317} \right] \text{ pb}.\tag{1.30}$$

At one-loop order, we consider the leading QCD correction to the inclusive hadronic cross section. This should not be confused with the complete one-loop electroweak correction to $e^+e^- \rightarrow \text{hadrons}$.

The relevant contributions are the virtual-gluon correction to the outgoing quark current and the real-emission process

$$e^+e^- \rightarrow q\bar{q}g.\tag{1.31}$$

The virtual and real contributions are separately infrared divergent, but their divergences cancel in the inclusive cross section.

Because QCD couples vectorially to the quarks, the same correction applies to the vector and axial-vector parts of the quark current in the massless limit. It therefore multiplies the photon contribution, the Z -boson contribution, and their interference in the same way. Using the result of Section 26.3, we obtain

$$\begin{aligned}\sigma_{\text{NLO QCD}}(s) &= \sigma_{\text{tree}}(s) \left(1 + \frac{3C_F\alpha_s(s)}{4\pi} \right) + \mathcal{O}(\alpha_s^2) \\ &= \sigma_{\text{tree}}(s) \left(1 + \frac{\alpha_s(s)}{\pi} \right) + \mathcal{O}(\alpha_s^2),\end{aligned}\tag{1.32}$$

where $C_F = 4/3$ and σ_{tree} contains the photon, Z -boson, and photon-- Z interference contributions calculated in part (a).

This result includes the complete NLO QCD correction in the massless inclusive approximation. It does not include the full one-loop electroweak correction, which would additionally require electroweak self-energy, vertex, photon-- Z mixing, box, counterterm, and real-photon contributions.

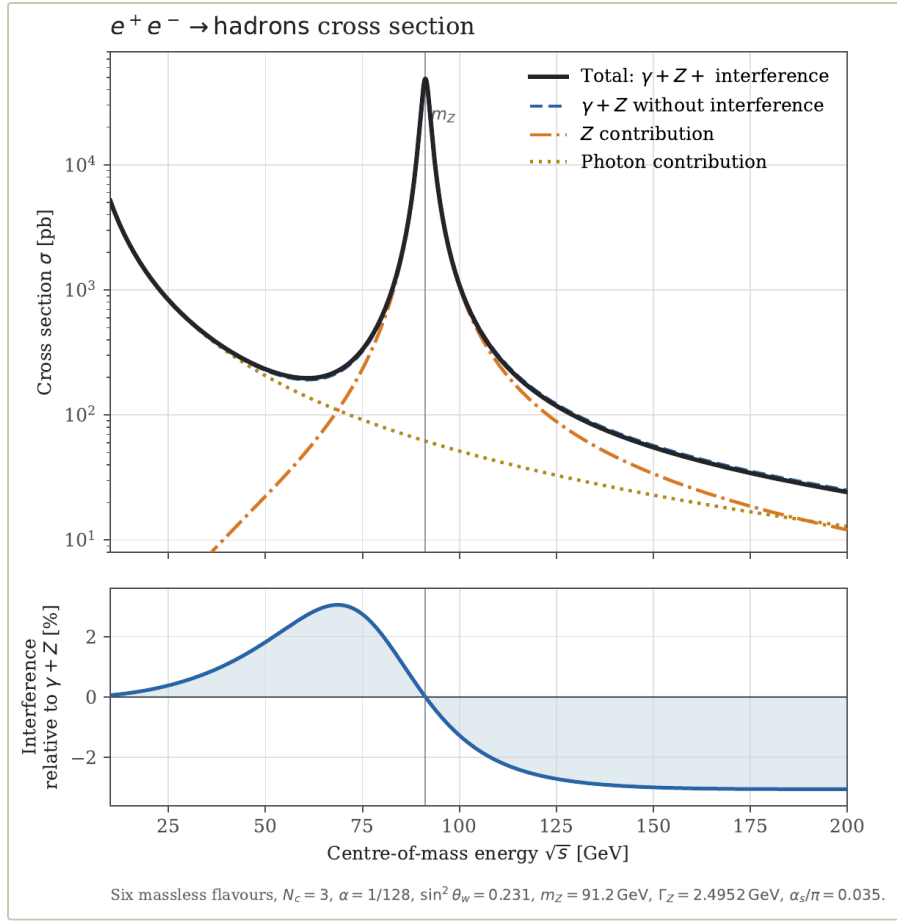


FIGURE 1. The NLO QCD cross section for $e^+e^- \rightarrow \text{hadrons}$, assuming six massless quark flavours and $N_c = 3$. The lower panel shows the interference contribution relative to the incoherent sum $\sigma_\gamma + \sigma_Z$.

The interference can be neglected when $s \ll m_Z^2$, where photon exchange dominates, and close to the Z resonance, where Z exchange dominates. In particular, the interference vanishes at $s = m_Z^2$ because it is proportional to $s - m_Z^2$. Away from these regions, it produces a correction of a few percent and should generally be retained.